🤖 AI Setup Logic: The Progression of a Trend

Mapping Al Brooks' Market Cycle directly to Python computational modeling.

> *"The market is always trying to break out, then trying to make every breakout fail. A breakout is a period of CERTAINTY — both sides agree current prices are wrong; the market moves quickly to find new uncertainty... The most important recurring pattern is the Spike and Channel, which eventually devolves into a Trading Range before reversing." — Al Brooks*

1 The Breakout (The Spike)

Opening Drive Spikes & High-Momentum Breakouts

BULL SPIKE BEAR SPIKE

The Cycle State: A period of absolute certainty. The market erupts from an equilibrium state (opening range or prior trading range). The goal here is to enter at the market or on a Stop order, as pullbacks are rare and shallow.

```
Baseline Volatility: avg_range = mean([last 20 bars range])
Momentum Threshold: 1.0x avg_range (Bars 1-4) or 1.25x avg_range (Bars 5+)
Algorithmic Trigger: Bar is a Strong Trend Bar AND range >= threshold
```

Note: To prevent the chaotic 9:30 AM print from blinding the moving average, the first 4 bars dynamically drop the strict scalar and accept any bar providing >50% of the opening volatility.

```
# algo_engine.py -> detect_opening_reversals_and_spikes()
threshold = 1.0 if i < 5 else 1.25
is_spike = bar.is_strong_bull(avg_range) and \
          (bar.range >= (avg_range * threshold) or (i < 5 and bar.range >= avg_ran

if is_spike:
    setups.append(Setup(
        setup_name="Spike / Opening Breakout",
        order_type="Stop/Market", # Buy the close of the strong spike
        confidence=0.60
    ))
```

2 The First Pullback (Transition to Channel)

High 1 / Low 1 Flags & Breakout Pullbacks

BULL FLAG (H1/H2) BEAR FLAG (L1/L2)

The Cycle State: The initial Spike is exhausted, and early profit-takers step in, creating the first pullback. This first pullback forms the start of the "Channel" phase. The algorithm natively tracks localized pullbacks to fade weak counter-trenders.

Pullback Definition: A bar with `close < open` (Bear Bar) OR `low < low[last 2 bars]`
Resumption Trigger: The exact bar that breaks `high > prev.high`
Grading Lookup: Count occurrences of the "Pullback Definition" over a rolling 5-bar window to assign H1 or H2.

```
# algo_engine.py -> detect_high_low_flags()
def is_pullback(b, prev, prev2):
    return b.is_bear or b.low < prev.low or b.low < prev2.low
```

```
if bar.high > prev.high and is_pullback(prev, bars[i-2], bars[i-3]):
    # Look back 5 bars to count total recent pullbacks
    num_pb = sum(1 for j in range(max(0, i-5), i)
                 if is_pullback(bars[j], bars[max(0, j-1)], bars[max(0, j-2)]))

    flag_num = min(3, max(1, num_pb))
    setups.append(Setup(name=f"High {flag_num} Bull Flag", order_type="Stop"))
```

3 The Trading Range (Equilibrium)

Fading Opening Reversals & Limit Order Traps

LIMIT ORDER SCALP | HIGH-PROBABILITY TRAP

The Cycle State: The Channel has lost its steepness and devolved into a Trading Range (uncertainty). In a trading range, *most breakout attempts fail*. The algorithm switches to fading weak breakouts using Limit Orders.

Trap Logic: Sequence of `[Massive Trend Bar] -> [Opposite Massive Trend Bar]`
Weak Pullback Logic: `range < ema * 0.002`

```
# user_algos/limit_setups.py
if is_spike_bull and prev_is_spike_bear:
     # The Bears spiked down, but Bulls immediately spiked it back up! The Bears a
     setups.append(Setup(
         setup_name="Opening Reversal (Bull Trap)",
         order_type="Stop/Limit", # Fade the weak bears
         confidence=0.70 # Exceedingly high probability layout
     ))
```

4 The Reversal

Major Trend Reversals (MTR)

TREND CHANGE

The Cycle State: The Trading Range has exhausted the dominant side. A Major Trend Reversal begins with a strong break of the Moving Average, followed by a failed re-test of the absolute extreme. A new Spike is about to begin in the opposite direction.

Trend Validation: A multi-leg push generating an absolute extreme (Swing High or Low).
MA Break: A strong close clearly crossing the 20-bar EMA.
Test & Failure: A subsequent push that tests the absolute extreme but fails to maintain the trend.

```
# user_algos/best_setups.py
abs_high = max(b.high for b in bars[max(0, i-30):i])
# Was the highest point at least 5 bars ago? (Significant pullback required)
if recent_high.idx <= bar.idx - 5:
    # Did we cross the EMA during the pullback?
    if min(b.low for b in bars[recent_high.idx:i]) < ema[i-2]:
        if bar.is_strong_bear(avg_range): # The re-test failed with a strong bear
            setups.append(Setup(name="Major Trend Reversal", order_type="Stop", co
```